

CareFlow

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Progress So Far

Our Careflow system is currently approximately 55% complete. So far, we have successfully implemented 11 use cases, focusing on core functionality such as clinic location management, patient and staff management, billing operations, and document handling.

Location Management
- Add new clinic locations
- Remove clinic locations
- Update clinic locations

Patient Management
- Add patient
- Staff Management
- Add staff

Billing System
- Add bills
- Pay bill

Records / Document Management
- Upload document
- Download document
- Delete document

Demo of Features



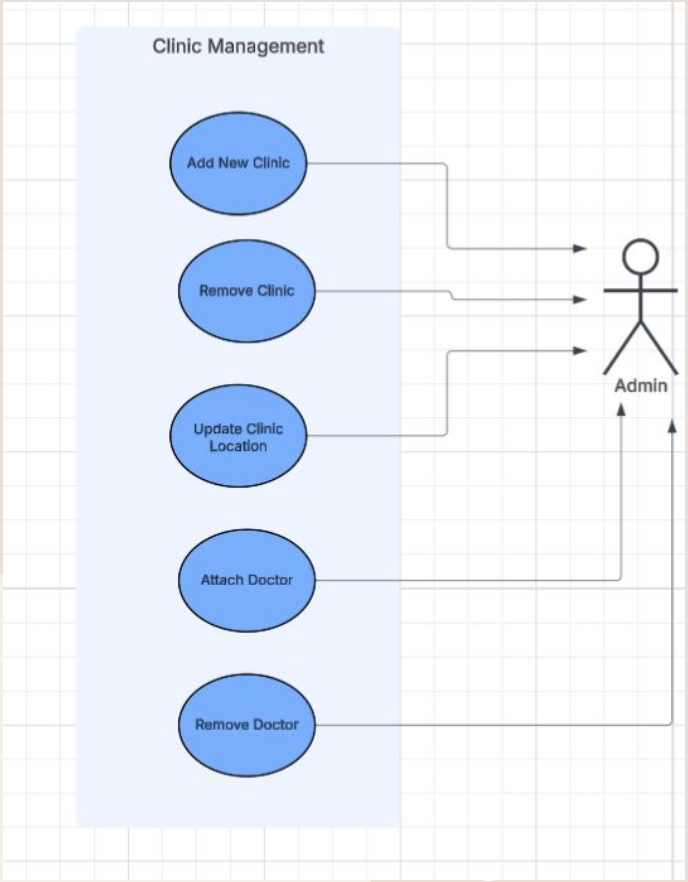
Use Case Tables

#	Use Case	SQL Query	Table	File	Description
1	**Upload Document**	`INSERT INTO records (patient_id, staff_id, filename, filepath, upload_date) VALUES (?, ?, ?, ?, ?)`	records	records.py	This allows users to upload a document file to the records directory. The button will pull up the file system, allowing the user to select the file to upload. The function will then take the file metadata and inserts it into the records table, tracking the filename, filepath and upload date, as well as tracking the patient id and staff id associated with the file.
2	**Download Document**	`SELECT filepath FROM records WHERE record_id = ?`	records	records.py	This allows users to download the file they have selected from the listbox. It finds the filepath using the table entry "filepath". The filesystem will open and allow them to determine where they want to save the file to.
3	**Delete Document**	`SELECT filepath FROM records WHERE record_id = ?` `DELETE FROM records WHERE record_id = ?`	records	records.py	This allows users to select a file from the listbox and delete it from the record directory and the file's entry and metadata from the records table.
4	**Search for Document**	`SELECT record_id, filename, upload_date FROM records WHERE patient_id = ? ORDER BY upload_date DESC`	records	record.py	This allows users to type a filename in the searchbox at the top of the page. As the user types in the name, the listbox will refresh and pull relevant file names from the records table.
5	**Add New Clinic Location**	`INSERT INTO ClinicLocation (name, address, city, state, zip, phone, status) VALUES (?, ?, ?, ?, ?, ?, 'active')`	ClinicLocation	Clinic_Location.py	This allows users to add a new clinic location and relevant data (name, address, city, zip_code, phone) to the ClinicLocation table. A pop up form will appear with relevant fields to input data.
6	**Update Clinic Location**	`UPDATE ClinicLocation SET {', '.join(fields)} WHERE location_id = ?`	ClinicLocation	Clinic_Location.py	This allows users to change any clinic location information they need to update. A pop form appears with the current data and the user can append these and save it to the ClinicLocation table.

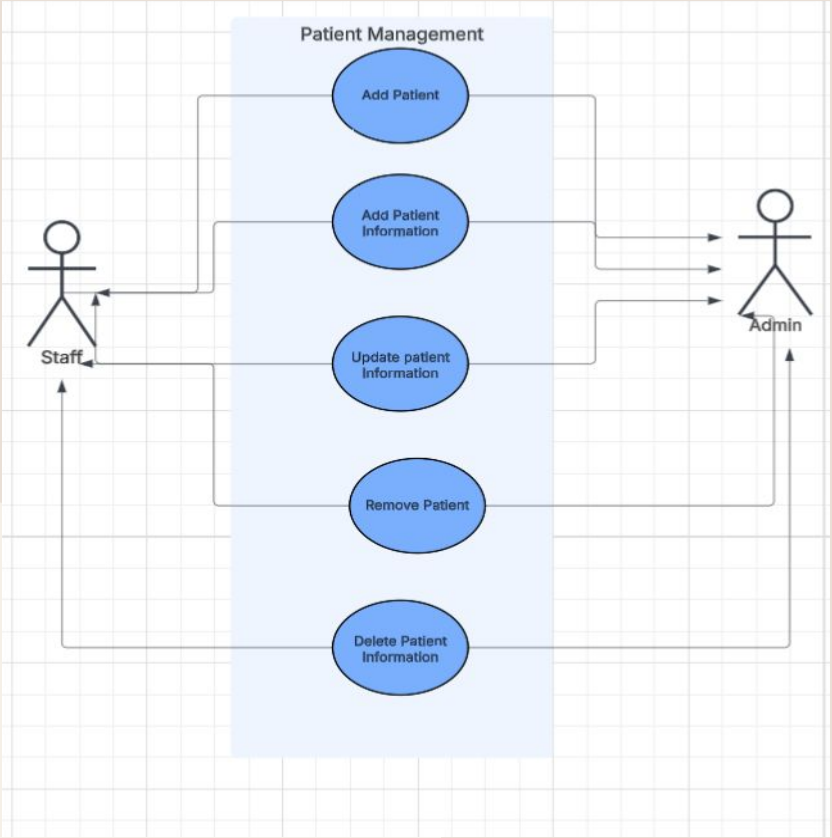
Use Case Tables

#	Use Case	SQL Query	Table	File	Description
7	**Delete/Remove Clinic Location**	`UPDATE ClinicLocation SET status = 'inactive' WHERE location_id = ?`	ClinicLocation	Clinic_Location.py	This allows users to delete the clinic entry from the database. This will reflect in the GUI when the listbox refreshes.
8	**Add Bills**	`INSERT INTO Bill (patient_id, location_id, amount, due_date, status) VALUES (?, ?, ?, ?, 'unpaid')`	Bill	Billing_Staff_app.py	This allows the user to create a new bill for a specific patient at a specific clinic. This also allows the user to set the amount the patient owes, as well as the due date.
9	**Pay Bill**	`UPDATE Bill SET status = 'paid' WHERE bill_id = ?`	Bill	Billing_Patient_app.py	This allows the user to pay a bill they have assigned to them through the patient ID. The user must first fill out a payment method to pay the bill.
10	**Add Patient**	`INSERT INTO Patient (first_name, last_name, dob, sex, phone, email, address, location_id, allergies, conditions, medications, notes, emergency_contact, password_hash) VALUES (?,?,?,?,?,?,?,?,?,?,?,?,?,?)`	Patient	Patient.py	The user can add a patient to the patient table. This will hold relevant patient info and history, such as name, date of birth, conditions, etc.
11	**Add Staff**	`INSERT INTO Staff (first_name, last_name, email, phone, role, active_flag, password_hash) VALUES (?, ?, ?, ?, ?, 1, ?)`	Staff	Staff.py	The user can add a new staff member to a clinic. It asks for first name, last name, email, phone number, role (doctor, nurse,etc) and password. This will save them to the staff table. This requires a special staff code to be added.

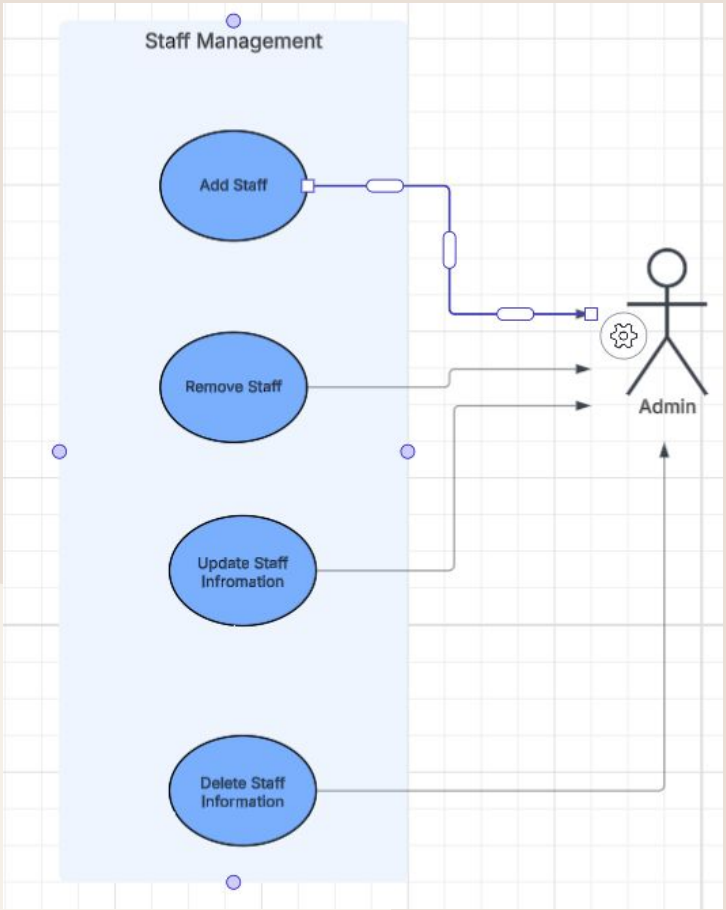
Clinic Management Use Cases



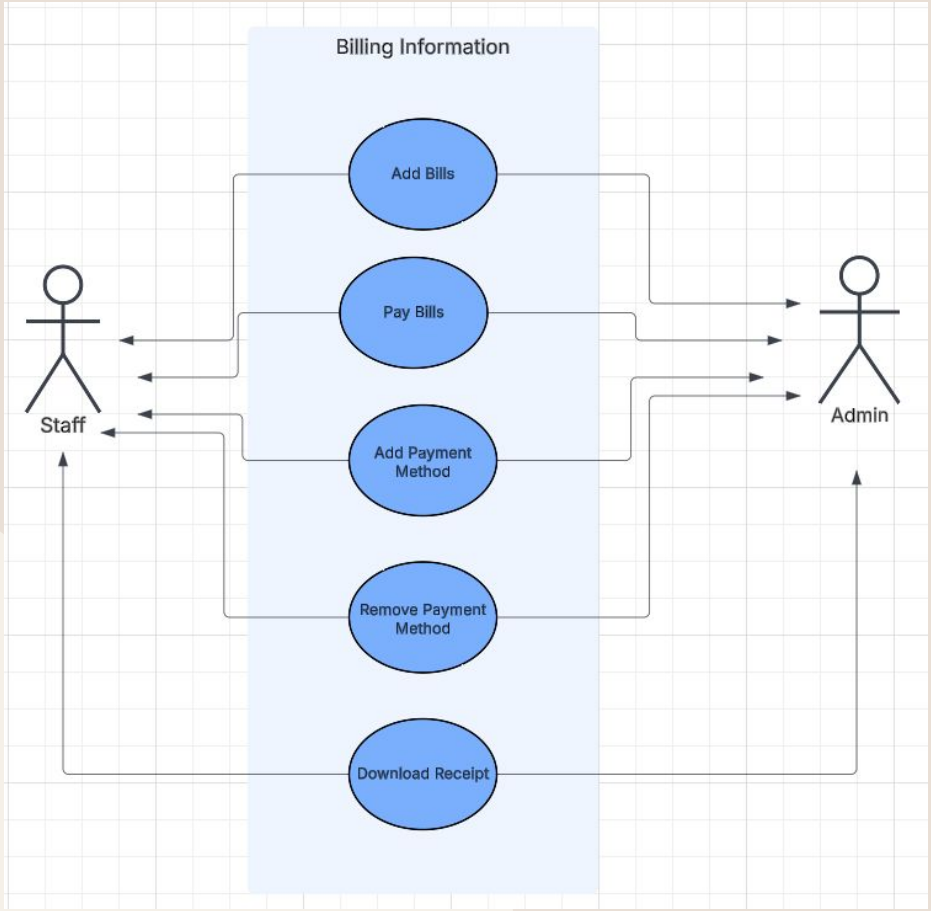
Patient Management Use Cases



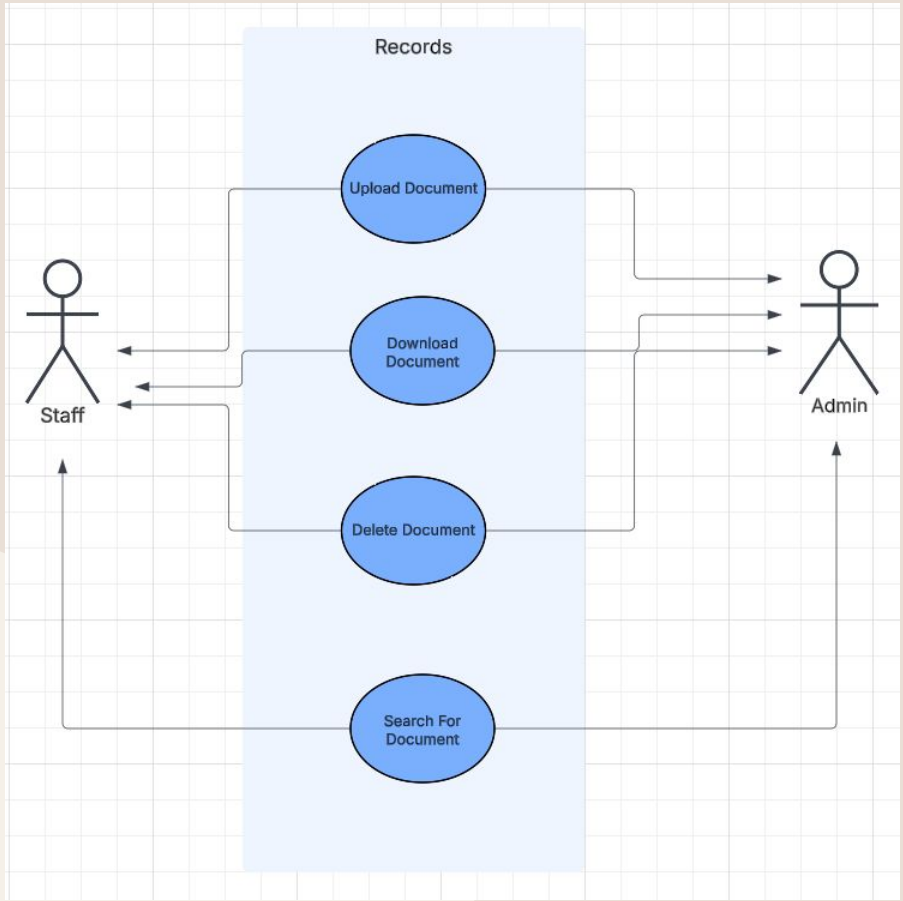
Staff Management Use Cases



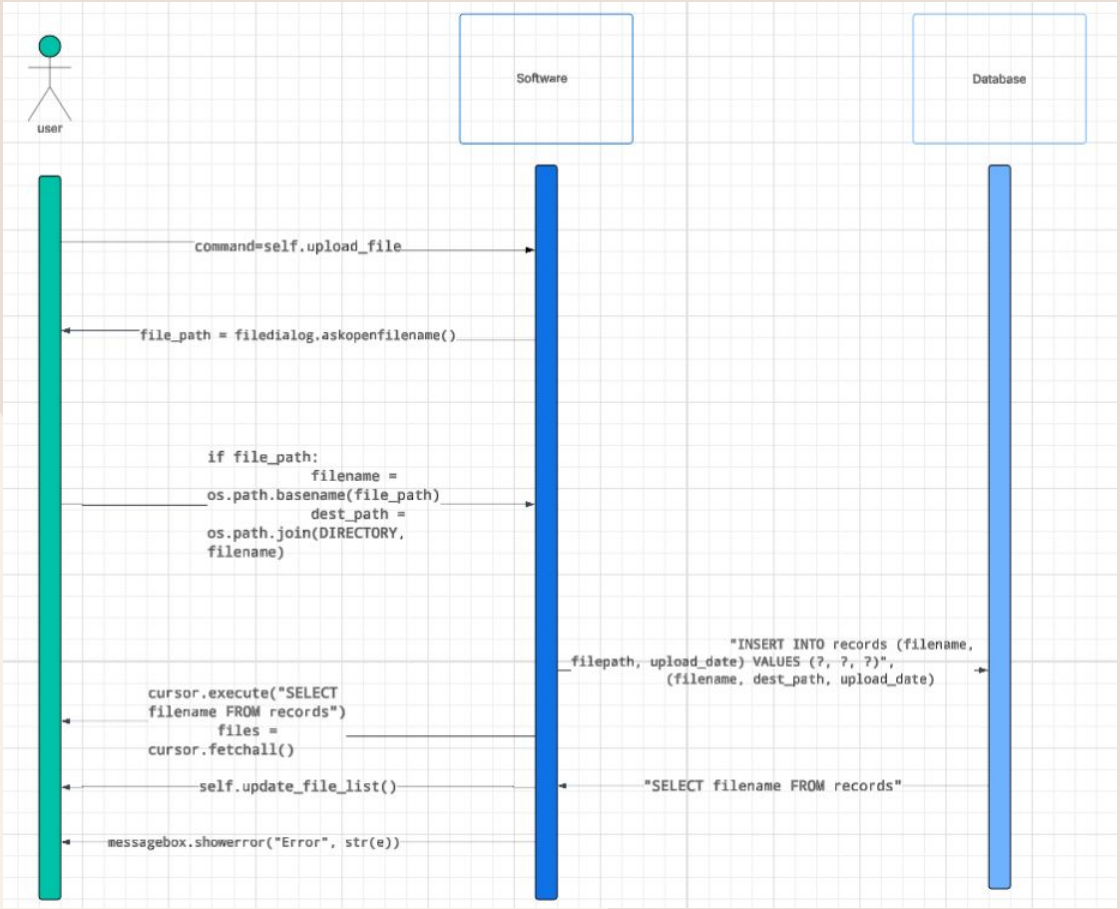
Billing Information Use Cases



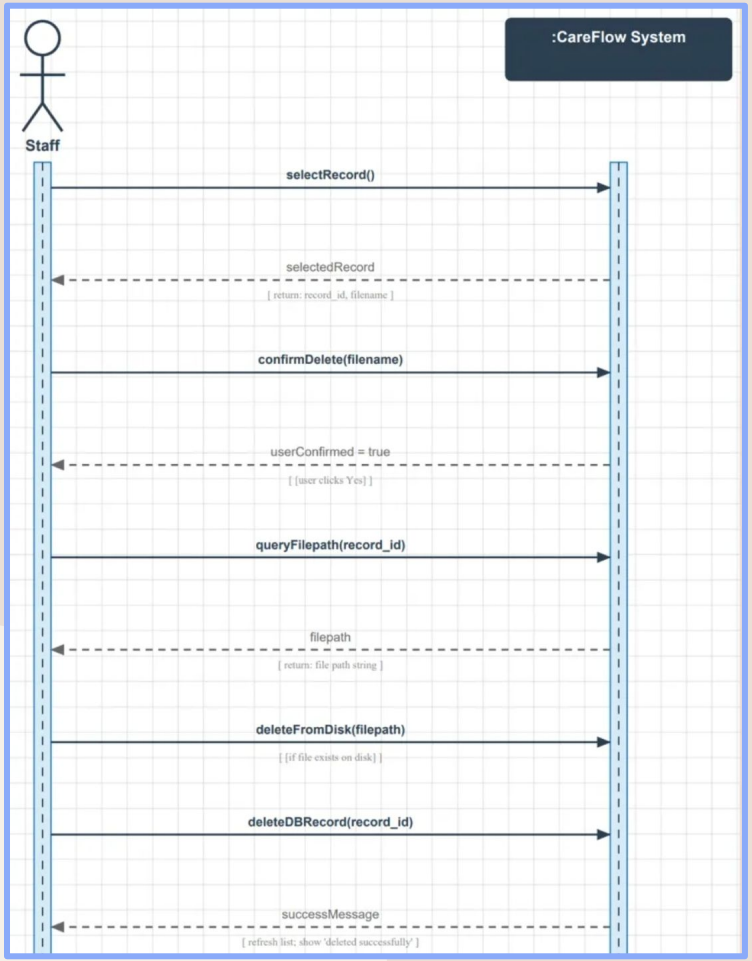
Records Use Cases



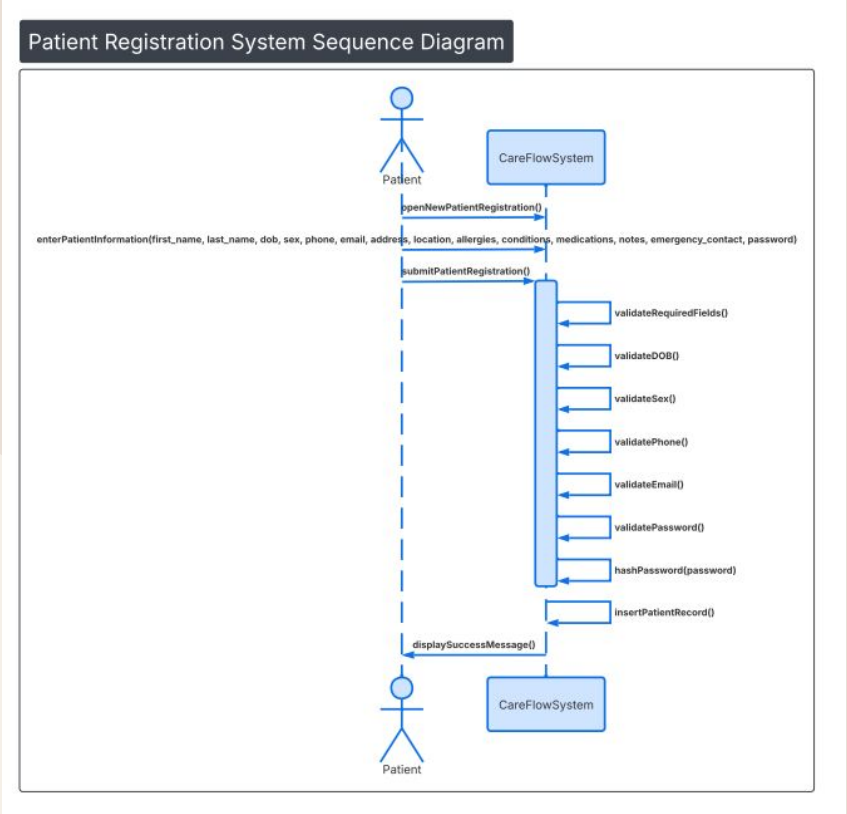
Upload File SSD (Zakaria Hoque)



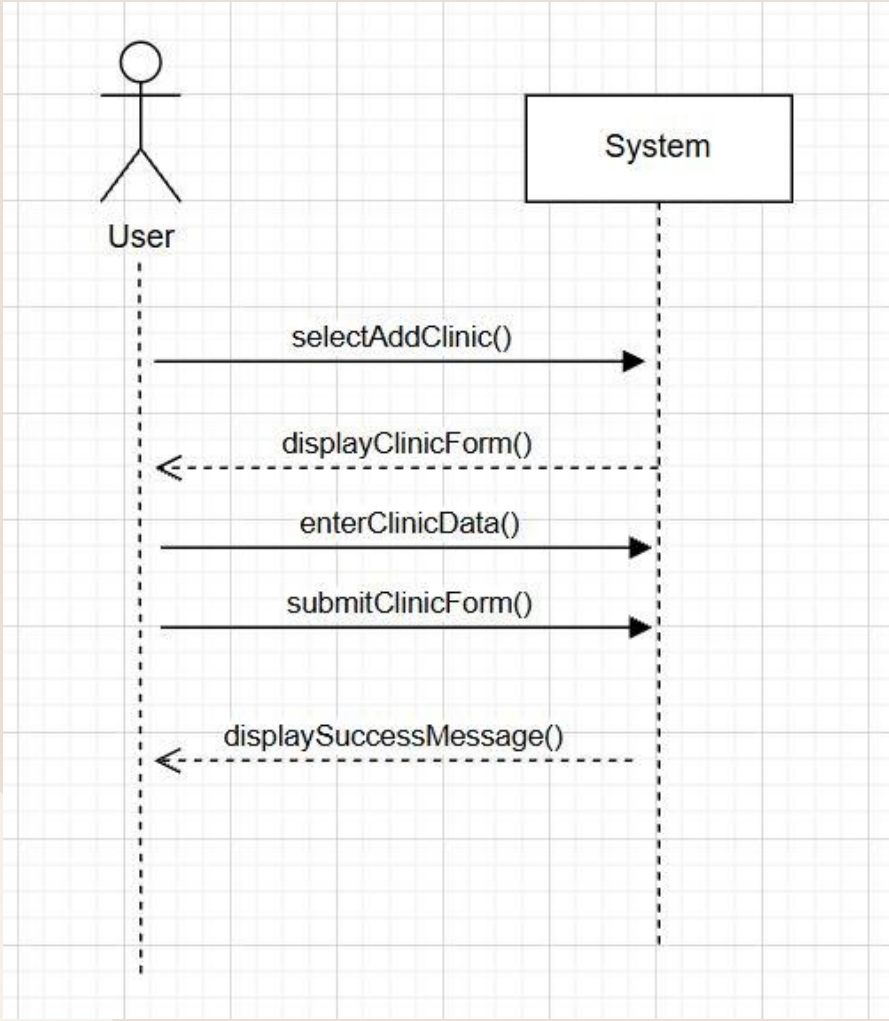
Delete File SSD (Manuel Gardea)



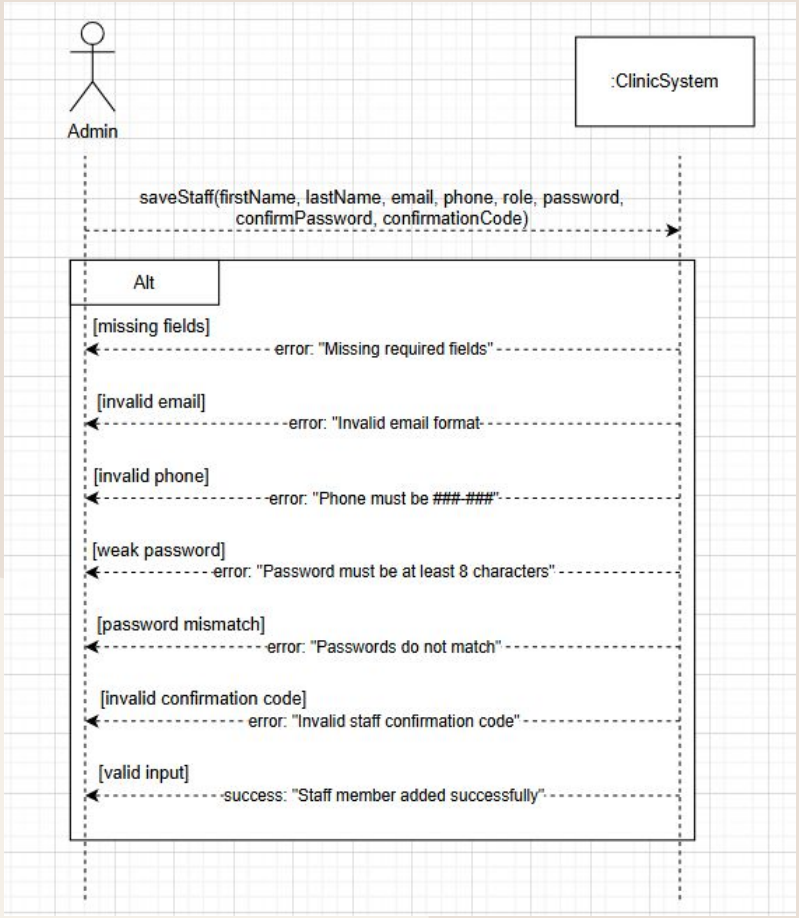
Add Patient SSD (Logan Templin - updated)



Add Clinic Location SSD (Tess Esman)



Add Staff SSD (Mara Evans)



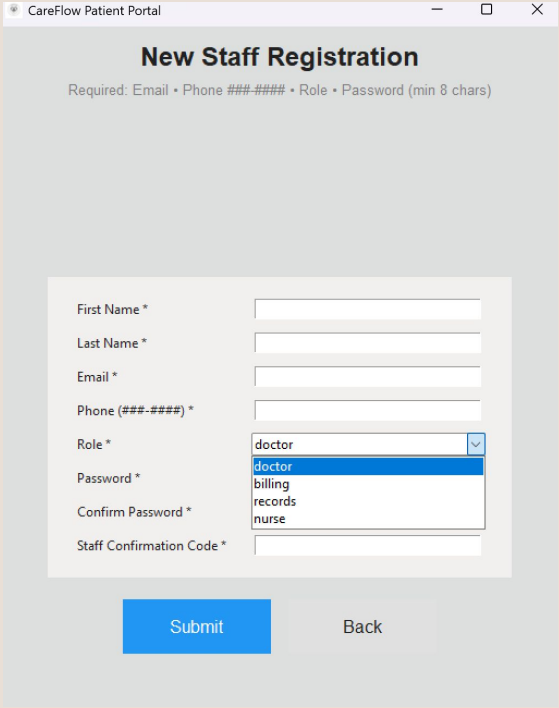
HCI Rule 5: Prevent Errors

Meaning:

- Interface should prevent serious errors
- If errors occur, the system should provide clear instruction for recovery

Application in CareFLow

- Implemented on the New Staff Registration.
- When adding a staff registration:
  - A Drop down list is provided to prevent user error in selecting a role as well as a confirmation code for certain role



HCI Rule 3: Offer Informative Feedback

Meaning:

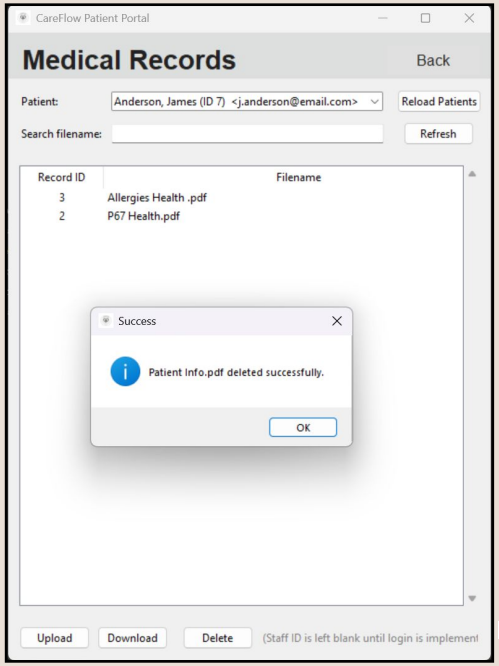
- Every user action should generate clear feedback.
- Important actions should provide stronger, visible responses.

Application in CareFLow

- Implemented on the Medical Records page.

Examples:

- Uploading a file → “[filename] uploaded successfully.”
- Deleting a record → “[filename] deleted successfully.”
- All actions (success or failure) trigger a message box response so users always know the outcome.



HCI Rule 6: Permit Easy Reversal of Actions

Meaning:

- Users should be able to undo or reverse actions whenever possible.
- This reduces anxiety and encourages exploration.

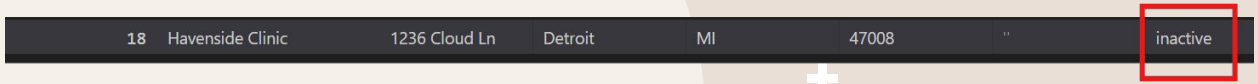
Application in CareFLow

- Implemented on the Clinic Locations page using soft delete.

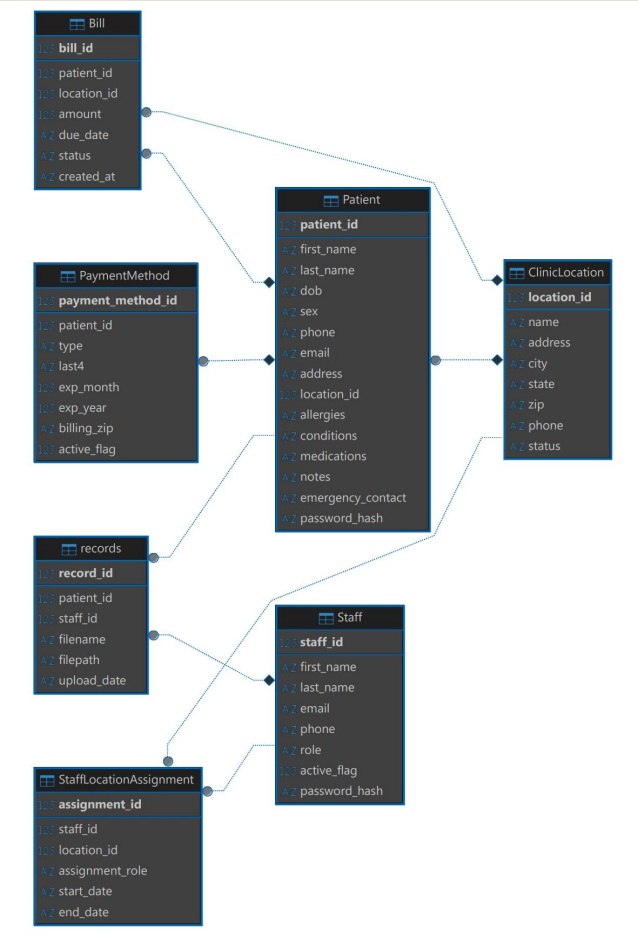
How it works:

- When a clinic is deleted
  - The system sets status = 'inactive' in the database.
  - The record is not permanently removed.
  - Deleted clinics can be restored if removed accidentally.

- Active clinics are displayed using:
  - get_all_active_clinics() → only shows status = 'active'



ERD


Bill
bill_id
patient_id
location_id
amount
due_date
status
created_at
Patient
patient_id
first_name
last_name
dob
sex
phone
email
address
location_id
allergies
conditions
medications
notes
emergency_contact
password_hash
PaymentMethod
payment_method_id
patient_id
type
last4
exp_month
exp_year
billing_zip
active_flag
ClinicLocation
location_id
name
address
city
state
zip
phone
status
records
record_id
patient_id
staff_id
filename
filepath
upload_date
Staff
staff_id
first_name
last_name
email
phone
role
active_flag
password_hash
StaffLocationAssignment
assignment_id
staff_id
location_id
assignment_role
start_date
end_date

Database Tables

ClinicLocation:

location_id	name	address	city	state	zip	phone	status
1	Main Clinic	123 Health St	Detroit	MI	48201	555-1111	active
2	West Clinic	456 Care Ave	Detroit	MI	48202	555-2222	active
3	Main Clinic	123 Health St	Detroit	MI	48201	555-1111	active
4	West Clinic	456 Care Ave	Detroit	MI	48202	555-2222	active
5	Northside Clinic	111 North Rd	Detroit	MI	48203	555-3001	active
6	South Health Center	222 South Blvd	Detroit	MI	48204	555-3002	active
7	East Medical	333 East Ave	Detroit	MI	48205	555-3003	active
8	West Care Center	444 West Dr	Detroit	MI	48206	555-3004	active
9	Lakeside Clinic	555 Lake Shore	Detroit	MI	48207	555-3005	active
10	Downtown Med Center	Detroit	MI	MI	48226	555-6789	inactive
11	Test Clinic	123 Test St	Detroit	MI	48201	555-1234	inactive
12	Havenside Clinic	1234 North St	Detroit	MI	48272	''	inactive
13	Havenside	1234 Cloud Ln	Detroit	MI	47007	''	inactive
14	Havenside Clinic	1234 Cloud Ln	Detroit	MI	47007	''	inactive
15	Havenside Clinic	1234 Cloud Ln	Detroit	MI	47007	''	inactive
16	Haven Medical Center	1234 Altlas Ln	Detroit	MI	47008	''	inactive
17	Downtown Medical Center	1234 Altas Ln	Detroit	MI	47006	''	active
18	Havenside Clinic	1236 Cloud Ln	Detroit	MI	47008	''	inactive
19	NorthEast Clinic	50345	Troy	MI	48043	123-1234	inactive
20	Troy Clinic	434 E	Detriot	MI	3214	833-3213	active
21	Field Clinic	Rochester AVE	Rochester	MI	4331	313-344	active
22	Favorite Health	32133 E ave	Macomb	MI	48023	312-3243	active
23	Fresh Health	3234 E Road	Oakland	MI	6753	232-3323	active
24	Enhanced Health	32131233 Road	Detriot	MI	3213	546-6789	active
25	Healthier Clinic	4321 E	Detriot	MI	4321	432-1323	active
26	Thirty Clinic	3121 E road	Red	MI	2123	545-2311	active
27	Twenty8 Clinic	Healthy Rd	Detriot	MI	41223	312-3213	active
28	First Health	Cloud LN	Detriot	MI	31312	313-3213	active
29	Last Health	3213 E road	Macomb	MI	3213	565-3133	active
30	MockUP Clinic	312321 E road	Macomb	MI	54321	321-3313	active

Bill:

bill_id	patie...	locati...	amount	due_date	status	created_at
1	1	1	120.5	2026-03-01	unpaid	2026-02-11 17:09:30
2	2	2	85	2026-03-05	paid	2026-02-11 17:09:30
3	3	3	200.75	2026-03-10	unpaid	2026-02-11 17:09:30
4	4	4	45.25	2026-03-12	paid	2026-02-11 17:09:30
5	5	5	310	2026-03-15	unpaid	2026-02-11 17:09:30
6	6	6	95.6	2026-03-18	unpaid	2026-02-11 17:09:30
7	7	7	150	2026-03-20	paid	2026-02-11 17:09:30
8	8	8	60.4	2026-03-22	unpaid	2026-02-11 17:09:30
9	9	9	500	2026-03-25	unpaid	2026-02-11 17:09:30
10	10	10	75.99	2026-03-28	paid	2026-02-11 17:09:30
11	5	5	40	2026-03-11	paid	2026-03-03 17:58:57

Staff:

record_id	patient_id	staff_id	filename	filepath	upload_date
2	7	NULL	P67 Health.pdf	record_files\P67 Health.pdf	2026-03-07 13:28:00
3	7	NULL	Allergies Health .pdf	record_files\Allergies Health .pdf	2026-03-07 13:28:07
4	4	NULL	Allergies Health _2.pdf	record_files\Allergies Health _2.pdf	2026-03-07 13:35:28
5	4	NULL	Patient Info_2.pdf	record_files\Patient Info_2.pdf	2026-03-07 13:35:38
6	8	NULL	Patient Info_3.pdf	record_files\Patient Info_3.pdf	2026-03-07 13:35:50
7	8	NULL	Allergies Health _3.pdf	record_files\Allergies Health _3.pdf	2026-03-07 13:35:59
8	5	NULL	Patient Info_4.pdf	record_files\Patient Info_4.pdf	2026-03-07 13:36:08
9	5	NULL	Allergies Health _4.pdf	record_files\Allergies Health _4.pdf	2026-03-07 13:36:17
10	10	NULL	Patient Info_5.pdf	record_files\Patient Info_5.pdf	2026-03-07 13:36:26
11	10	NULL	Allergies Health _5.pdf	record_files\Allergies Health _5.pdf	2026-03-07 13:36:33
12	3	NULL	Patient Info_6.pdf	record_files\Patient Info_6.pdf	2026-03-07 13:36:53
13	3	NULL	Allergies Health _6.pdf	record_files\Allergies Health _6.pdf	2026-03-07 13:37:00

Database Tables

Patient:

patient_id	first_name	last_name	dob	sex	phone	email	address	locati...	allergies	conditions	medicati...	notes
1	John	Doe	1990-05-10	M	555-1111	john.doe@email.com	101 Main St	1	Peanuts	Asthma	Albuterol	Prefers morning visits
2	Jane	Smith	1985-03-22	F	555-2222	jane.smith@email.com	202 Oak Ave	1	Penicillin	Diabetes	Metformin	Needs wheelchair access
3	Michael	Brown	1978-11-02	M	555-3333	mike.b@email.com	303 Pine Rd	1	None	High blood pressure	Lisinopril	Monitor BP
4	Emily	Davis	1995-07-18	F	555-4444	emily.d@email.com	404 Cedar St	1	Shellfish	None	None	First-time patient
5	Daniel	Wilson	1988-01-30	M	555-5555	dan.w@email.com	505 Birch Ln	1	Latex	Eczema	Hydrocortisone	Skin sensitivity
6	Olivia	Taylor	2001-09-12	F	555-6666	olivia.t@email.com	606 Maple Dr	1	None	Migraines	Ibuprofen	Sensitive to light
7	James	Anderson	1969-04-25	M	555-7777	j.anderson@email.com	707 Lake Rd	1	Dust	Arthritis	Naproxen	Chronic joint pain
8	Sophia	Thomas	1999-12-15	F	555-8888	sophia.t@email.com	808 River St	1	Gluten	Celiac disease	Dietary control	Strict diet
9	William	Jackson	1975-06-08	M	555-9999	will.j@email.com	909 Hill Ave	1	Bee stings	None	EpiPen	Carries EpiPen
10	Ava	White	1993-02-27	F	555-0000	ava.w@email.com	1000 Elm St	1	Cats	Anxiety	Sertraline	Prefers telehealth

Records:

record_id	patient_id	staff_id	filename	filepath	upload_date
2	7	NULL	P67 Health.pdf	record_files\P67 Health.pdf	2026-03-07 13:28:00
3	7	NULL	Allergies Health .pdf	record_files\Allergies Health .pdf	2026-03-07 13:28:07
4	4	NULL	Allergies Health _2.pdf	record_files\Allergies Health _2.pdf	2026-03-07 13:35:28
5	4	NULL	Patient Info_2.pdf	record_files\Patient Info_2.pdf	2026-03-07 13:35:38
6	8	NULL	Patient Info_3.pdf	record_files\Patient Info_3.pdf	2026-03-07 13:35:50
7	8	NULL	Allergies Health _3.pdf	record_files\Allergies Health _3.pdf	2026-03-07 13:35:59
8	5	NULL	Patient Info_4.pdf	record_files\Patient Info_4.pdf	2026-03-07 13:36:08
9	5	NULL	Allergies Health _4.pdf	record_files\Allergies Health _4.pdf	2026-03-07 13:36:17
10	10	NULL	Patient Info_5.pdf	record_files\Patient Info_5.pdf	2026-03-07 13:36:26
11	10	NULL	Allergies Health _5.pdf	record_files\Allergies Health _5.pdf	2026-03-07 13:36:33
12	3	NULL	Patient Info_6.pdf	record_files\Patient Info_6.pdf	2026-03-07 13:36:53
13	3	NULL	Allergies Health _6.pdf	record_files\Allergies Health _6.pdf	2026-03-07 13:37:00

PaymentMethod:

payment_method_id	patient_id	type	last4	exp_mon...	exp_year	billing_zip	active_flag
1	1	card	1234	5	2027	48201	1
2	2	card	2345	6	2026	48202	1
3	3	bank	3456	7	2028	48203	1
4	4	card	4567	8	2027	48204	1
5	5	card	5678	9	2029	48205	1
6	6	bank	6789	10	2026	48206	1
7	7	card	7890	11	2027	48207	1
8	8	card	8901	12	2028	48226	1
9	9	bank	9012	1	2029	48210	1
10	10	card	0123	2	2027	48211	1

StaffLocationAssignment:

assignment_id	staff_id	location_id	assignment...	start_date	end_date
1	1	1	Administrator	2024-01-01	NULL
2	2	2	Billing Manager	2024-02-01	NULL
3	3	3	Records Clerk	2024-03-01	NULL
4	4	4	Patient Coordinator	2024-04-01	NULL
5	5	5	Billing Specialist	2024-05-01	NULL
6	6	6	Front Desk	2024-06-01	NULL
7	7	7	Admin Assistant	2024-07-01	NULL
8	8	8	Records Manager	2024-08-01	NULL
9	9	9	Patient Support	2024-09-01	NULL
10	10	10	Clinic Manager	2024-10-01	NULL